



Analysis of Ethnic Homophily in International Trade Using Large-Scale Surname Data

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Abstract

Although previous studies described the role of ethnic factors in international trade, due to the difficulty of estimating the ethnicity of trade entities, such analysis was limited to a few ethnic groups, and the differences in the strength of factors across ethnic groups were not identified. By estimating corporate ethnicity using the large-scale surname data of corporate managers, we quantitatively compare and analyze the dependence of various ethnic groups on ethnicity. Asian and Middle Eastern ethnic groups have strong ethnic homophily. An analysis of ethnic factors in commerce between same-language countries using the gravity model suggests that the effect of ethnicity is significant even when language barriers are removed.

Keywords International trade · Ethnicity · Ethnic homophily · Gravity model

1 Introduction

The role of ethnic networks is important in international trade, which is composed of diverse entities. Ethnic networks effectively reduce the cost of information barriers and unreliable contracts in such trade. Through empirical research, Rauch et al. [1] found that ethnic Chinese networks have greatly increased bilateral trade with markets in Southeast Asia. Other studies argued that the effects of ethnic Chinese networks have promoted bilateral trade [2, 3]. An important role in bilateral trade has also been found in other ethnic networks. In Africa, such networks increase bilateral trade between countries [4–7]. This trend appears firmer in corrupt or distrustful countries [4]. Felbermayr et al. clarified the role

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of ethnic networks in trade by expanding the scope of the ethnic groups studied to Morocco, Turkey, and Mexico [2].

Previous studies found that ethnic factors affect cross-country trade, although their scope was limited to a single ethnic group. This is because classifying ethnic groups and collecting data are complicated tasks. The ethnic networks in many related studies have been examined based on immigration data. The impact of ethnic networks has been analyzed using the statistical data of immigrants who emigrated from their birth country [1–3, 8]. If the composition of ethnic groups is too complex and obtaining statistical data is difficult, a study's analysis may be based on research datasets or survey data [4–7]. However, this remains unreliable because disentangling ethnic groups depends on the social environment.

Classifying ethnicity is challenging because it is affected by such factors as race, culture, language, and region. Therefore, race and cultural homogeneity are difficult to consider even when their classification is done for a single country. Moreover, addressing ethnic diversity in multi-ethnic countries is also complicated.

To address this problem, we use surname data to obtain ethnicity. Surnames are influenced by ethnicity, culture, and region. Due to these characteristics, surname data are actively used in research to estimate race and ethnicity [9–11]. In addition, the ethnic factors estimated by surnames have been applied in a wide range of fields, including sociology [12], demographics [13], and biomedicine [14].

We analyze ethnic homophily in international trade using the surnames of those in managerial positions of their companies based on the surname-origin classifier developed in our previous study [15]. This classifier predicts ethnicity based on surnames. With the classified results, we investigate a company's majority ethnic groups working in managerial positions, calculate the transaction frequency among ethnicities to find ethnic homophily in trade, and measure its statistical significance using the gravity model.

Since Jan Tinbergen [20] introduced the original gravity equation of bilateral trade, numerous empirical studies using the gravity model have been reported. Some of these empirical studies have proven that bilateral trade follows the gravity equation regardless of the country [21, 22]. The flows of bilateral trade are proportional to the size of the economy measured by each country's GDP and are inversely proportional to distance. Here, distance includes factors such as regulation, membership, and culture, rather than physical distance. In this study, we verify the effect of ethnic distance on bilateral trade using a gravity model.

This study offers the following contributions. We quantitatively compare and analyze the dependence on ethnicity of each ethnic group, rather than a single ethnic group, using surname data. Furthermore, we determine which ethnic groups are heavily influenced by ethnicity in trade.

The remainder of the paper is organized as follows. Section 2 introduces our dataset, and Sect. 3 explains the method of ethnicity classification and defines the ethnic groups. In Sect. 4, we show ethnic homophily in international trade by examining the relevant activity between ethnic groups. In Sect. 5, we measure the statistical significance of ethnicity factors in international trade using the gravity model. Section 6 compares domestic and international trade. We conclude in Sect. 7.

2 Dataset

2.1 FactSet Shipping Dataset

This dataset, from the customs clearance records at US ports provided by FactSet [16], contains all import transactions through American ports, shipping container by shipping container. This study analyzes data on import transactions from 2013 to 2018. Since this dataset also includes information on the people in managerial positions at the import and export companies, the country of origin of their surnames can be estimated using the methods described in Section 3. We explicitly investigate the dependence on ethnicity in the choice of international trading partners with the transaction data.

We removed self-loop transactions and unclear ownership of importing and exporting to clarify the relationship between exporter and importer. Accordingly, we excluded (1) transactions between parent-subsidiary relationships, (2) transactions in which the exporting company's nationality and the exporting port's locations do not match, and (3) transactions recorded as imports by financial companies making importing payments. In addition, we selected companies that have more than three managers for estimating the majority's ethnicity. After data cleaning, a total of 15,503,150 transactions are used, representing 27.7% of all transaction data.

2.2 FactSet Supply Chain Dataset

This dataset is FactSet's trade relationship information on 31,600 global companies. This information was collected from company annual reports and public news articles worldwide. Companies and people in managerial positions are linked in the dataset. The managerial positions include both low-level and top-level managers—for example, a team leader is a low-level manager, while a president or CEO is a top-level manager. Here, we call an employee in managerial positions a 'manager.' We selected companies that have more than three managers and removed the parent-subsidiary relationships. The cleaned dataset includes 479,448 pairs of supplier-customer relationships from 2014 to 2020.

2.3 WTFC Trade Dataset

The World Trade Flows Characterization (WTFC) dataset (provided by CEPII [17]) contains the trade values (in US dollars) of products in more than 230 countries, their GDPs, and the distance between cities with the highest population distribution of each country. We used the 2016 aggregated trade values by country, GDP, and distance.

2.4 ORBIS Businesspeople Dataset

The 2016 ORBIS dataset [18], which includes information on about 37 million company executives and stakeholders in 77 countries, was provided by Bureau van

Dijk. Using its contents as training data, we created the surname-origin classifier explained in Section 3, with which we determined the estimated ethnic composition of the managers in each country.

3 Surname-Origin Classifier

We used our surname-origin classifier [15] built using a large-scale surname dataset and machine learning. The surname-origin classifier takes a family name as an input value to predict its country of origin from 77 countries. The level of classification accuracy depends on the country (Table 1). The results for such single-ethnicity countries as Japan and the Republic of Korea have high accuracy, while those of such multi-ethnic countries as the US and Brazil have low accuracy.

We improved the classification accuracy by clustering 77 countries with a surname vector defined by the average of the log-likelihoods obtained when classifying each surname. We used k-means clustering, where k is determined by the silhouette method [19]. These 77 countries were clustered into $k=35$ ethnic groups (Table 1). Each ethnicity group is identified by the ISO code of the country with the most businesspeople (managers) who belong to it.

Multi-ethnic countries with a similar distribution of surname origins, such as the U.S. and the U.K., are grouped into the same ethnic group. The classification accuracy for ethnic group units is high, e.g., 0.606 for the GB group. We determined a company's majority ethnic group from the surnames of its managers and investigated the impact of ethnicity on international trade between companies.

4 Ethnic Homophily in International Trade

We determined the majority ethnic group for each of the 662,742 companies that contain information on over three managers. For example, companies with more than a majority of managers classified as Japanese are labeled as the JP group.

To investigate commerce between ethnic groups, we classified the importing and exporting companies into 16 ethnic groups (Table 1). We calculated the conditional probability of trade transactions to visualize the significance of transactions made between identical ethnic groups. The conditional probability of trade between ethnic groups is calculated as

$$P(T) = P(A|B)/P(A), \quad (1)$$

where $P(A|B)$ is the conditional probability that group B imports from group A and $P(A)$ is the unconditional probability of importing from group A.

We confirmed the independence of the unconditional probability of trading and its conditional probability by a z -test. After that, we expressed in a heatmap the value of $P(T)$, which was verified as independent ($p < 0.05$). Yellow denotes that there is no significant number of transactions between ethnic groups, while red

Table 1 Clustering results of surname origins

Ethnic groups	Countries
GB (0.606)	Australia (0.087), Canada (0.010), Ireland (0.488), New Zealand (0.001), South Africa (0.299), United Kingdom (0.544), United States (0.003)
AT (0.456)	Austria (0.454), Belgium (0.235), Netherlands (0.454)
CN (0.675)	China (0.672), Hong Kong (0.526), Malaysia (0.368), Singapore (0.136)
IT (0.690)	Italy (0.690)
DE (0.382)	France (0.316), Germany (0.333), Luxembourg (0.093), Switzerland (0.234)
IN (0.433)	Afghanistan (0.026), India (0.549), Indonesia (0.427), Israel (0.402), Papua New Guinea (0.280)
CO (0.562)	Colombia (0.461), Mexico (0.362), Venezuela (0.225)
BR (0.504)	Argentina (0.005), Brazil (0.358), Chile (0.366), Philippines (0.170), Portugal (0.393), Spain (0.165)
JP (0.874)	Japan (0.874)
SE (0.647)	Denmark (0.501), Norway (0.563), Sweden (0.417)
TW (0.632)	Taiwan (0.632)
AE (0.513)	Iraq (0.229), Kuwait (0.338), Saudi Arabia (0.374), Sudan (0.303), United Arab Emirates (0.457)
EG (0.592)	Egypt (0.269), Islamic Republic of Iran (0.532), Lebanon (0.480), Pakistan (0.634), Syrian Arab Republic (0.113)
KR (0.901)	Republic of Korea (0.901)
TR (0.672)	Turkey (0.672)
TH (0.787)	Thailand (0.787)
UA (0.417)	Ukraine (0.417)
KZ (0.607)	Kazakhstan (0.607)
KE (0.505)	Kenya (0.505)
PL (0.665)	Poland (0.697), Czech Republic (0.453), Slovakia (0.212), Slovenia (0.471)
DZ (0.592)	Algeria (0.604), Morocco (0.273)
GR (0.653)	Greece (0.653)
RO (0.691)	Romania (0.563), Albania (0.656), Hungary (0.623), Republic of Moldova (0.409)
VN (0.883)	Vietnam (0.883)
IS (0.921)	Iceland (0.921)
RS (0.822)	Serbia (0.822)
BG (0.755)	Bulgaria (0.755)
MK (0.726)	Republic of North Macedonia (0.726)
RU (0.605)	Russian Federation (0.605)
LT (0.770)	Lithuania (0.770)
FI (0.710)	Finland (0.710)
NG (0.614)	Nigeria (0.614)
LV (0.759)	Latvia (0.759)
CY (0.652)	Cyprus (0.652)
YE (0.497)	Yemen (0.497)

Numbers in parentheses refer to classification accuracy. In the FactSet shipping dataset, the upper 16 groups (from GB to TH) account for 97% of the total transaction records

denotes a significantly high number of transactions. A significantly low number of transactions is marked in green.

Figure 1a shows how actively ethnic groups in the US trade with fellow members of the same ethnic group. Ethnic homophily is observed in most ethnic groups. The Asian JP, TW, KR, TH and Middle Eastern AE, EG, TR ethnic groups have especially strong tendencies to trade with those in the same ethnic group.

Such ethnic homophily in international trade is also commonly observed in interfirm transactions with non-US countries. Figure 1b, c represent ethnicity-based activities in international trade with countries that belong to the DE and CN groups. For these countries, unlike the US, the dataset does not cover all shipping records, although we identified a trend of active transactions between identical ethnic groups.

5 Measuring Ethnic Factors in International Trade

We statistically investigated the contribution of ethnic connections to international trade using the gravity model. The standard gravity model for trade describes the amount of bilateral trade flow in terms of economic size and distance. We measured the contribution of ethnic factors by incorporating them in the gravity model. In our gravity model, we adopt a negative binomial pseudo maximum likelihood (NBPML) estimator.

We use the FactSet shipping and WTFC trade datasets in this section. The GDP and distance values for each country are taken from CEPII's gravity dataset.

First, we used the FactSet shipping dataset, which provides the ethnic groups of each company in the US. By investigating the statistical significance of the ethnic factors in US international trading, we can identify the differences by ethnicity. Next, we analyzed the statistical significance of the ethnic factors on a global scale from the WTFC trade dataset and checked whether ethnicity is statistically significant even when language barriers are removed.

5.1 Gravity Model

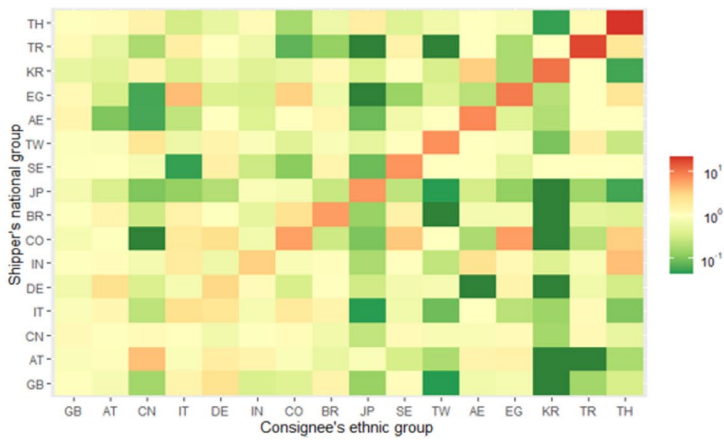
The standard gravity model can be represented as follows:

$$\ln F_{ij} = \beta_0 + \beta_1 \ln M_i + \beta_2 \ln M_j + \beta_3 \ln D_{ij} + \varepsilon_{ij}, \quad (2)$$

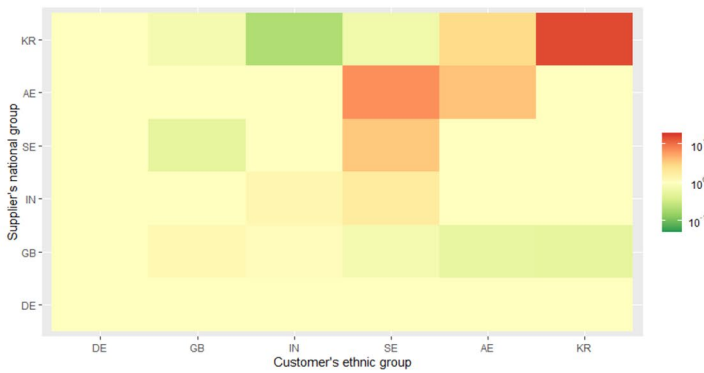
where i and j denote exporting and importing countries, respectively. F_{ij} is the volume of trade in US dollars between countries i and j , and M_i and M_j are their GDPs in US dollars. D_{ij} is the distance measured based on the most population-dense city of each country, and ε_{ij} is an error term.

We added ethnic factor E_{ij} to this standard gravity model:

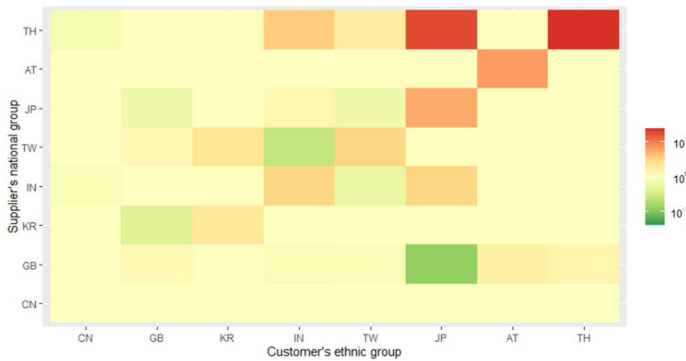
$$\ln F_{ij} = \beta_0 + \beta_1 \ln M_i + \beta_2 \ln M_j + \beta_3 \ln D_{ij} + \beta_4 E_{ij} + \varepsilon_{ij}, \quad (3)$$



(a)



(b)



(c)

Fig. 1 Frequency of transactions between ethnic groups: **a** consignees in US, **b** companies belonging to DE group, and **c** companies belonging to CN group

where E_{ij} represents the cosine similarity in the ethnic distribution between countries i and j :

$$E_{ij} = \frac{\mathbf{I} \cdot \mathbf{J}}{\|\mathbf{I}\| \|\mathbf{J}\|} = \frac{\sum_{k=1}^n I_k J_k}{\sqrt{\sum_{k=1}^n I_k^2} \sqrt{\sum_{k=1}^n J_k^2}},$$

where vectors $\mathbf{I}$ and $\mathbf{J}$ are the ethnic compositions of countries i and j . I_k is a component of vector $\mathbf{I}$, and $n=35$ is the number of ethnic groups. The ethnic distribution (vector) of corporate managers in each country is obtained from the ORBIS dataset. For example, the cosine similarity between the ethnic composition of the US and the UK is 0.9772, while this value for the US and China is 0.0565.

The gravity model of trade between country i and ethnic group k within country j can be described by extending Eq. (3):

$$\ln F_{i, k \in j} = \beta_0 + \beta_1 \ln M_i + \beta_2 \ln M_{k \in j} + \beta_3 \ln D_{ij} + \beta_4 E_{i, k \in j} + \varepsilon_{i, k \in j}, \quad (4)$$

where $F_{i, k \in j}$ is the volume of trade in US dollars between country i and ethnic group k within country j . We assume the following GDP of ethnic group k in country j : $M_{k \in j}$ = (rate of company's majority ethnic group k in country j)/(total GDP of country j). $E_{i, k \in j}$ represents the similarity of the ethnic composition between ethnic group k within countries j and i . The statistical significance of the ethnic factor in international trade is clarified by model selection based on Akaike's Information Criterion (AIC) and regressions on the three gravity models expressed by Eqs. (2, 3, 4).

5.2 Statistical Significance of Ethnic Factor in International Trade

Using the gravity models introduced in the previous section, we show that ethnic factors are statistically significant in international trade. In particular, we confirmed their significance for both imports and exports with Asian countries. In the multi-ethnic United States, companies in Asian ethnic groups tend to choose Asian countries as trading partners, while companies in non-Asian ethnic groups avoid choosing specific countries. Such bias of ethnicity is thus even confirmed among ethnic groups that speak a common language.

We investigated whether the gravity model with/without an ethnic factor (Eqs. 2 or 3) is chosen based on AIC for all combinations of countries included in the WTFC trade dataset. We selected the gravity model with an ethnic factor, which is statistically significant (Table 2).

We clarified the country dependence of the ethnic factor's significance by fixing exporting country i or importing country j in the gravity models. Here, we use both the WTFC trade dataset and the FactSet shipping dataset, which shows the number of shipping containers imported by the United States. Table 3 shows the regression coefficients and the AIC of the gravity models when importing country j is fixed. China and Korea tend to import from countries with a similar ethnic composition. In other countries, the gravity model with an ethnic factor is not supported by the AIC. We next

Table 2 Regression coefficients and AIC are shown for gravity models with/without ethnic factors (Eqs. 2, 3) calculated for all combinations of exporting and importing countries recorded in WTCF trade dataset

Gravity model	Constant β_0	GDP of exporter β_1	GDP of importer β_2	Distance β_3	Ethnicity β_4	AIC
Without ethnic factor	– 13.36***	0.91***	0.86***	– 0.98***		93245
With ethnic factor	– 13.47***	0.91***	0.85***	– 0.94***	0.51***	93216

*** p value < 0.001, ** p value < 0.01, * p value < 0.05, p value < 0.1

display in Table 4 the regression coefficients and the AIC of the gravity models when exporting country i is fixed. Regarding exports, as seen for imports above, China and Korea export to countries with a similar ethnic composition. Japan and Spain show similar features. On the other hand, no significant ethnic effect is observed in exports from the US, Germany, and Brazil.

As shown in Tables 3, 4, the US and Europe do not tend to choose countries with a similar ethnic composition as trading partners, reflecting a feature of their overall multi-cultural societies. However, companies in Asian ethnic groups, which are minorities in US and Europe, often do choose countries of identical ethnicity as trading partners. Table 5 shows the regression coefficients and the AIC of the gravity model (Eq. 4) for each ethnic group in the United States. In ethnic groups CN, JP, TW, EG, KR, and TH, the gravity model incorporating ethnic factors is selected with a lower AIC score, and the factor's coefficient is statistically significant. These results are consistent with the matrix of trade frequencies among ethnic groups in Fig. 1. The active commerce among identical ethnic groups in Fig. 1, which cannot be explained by GDP and distance, shows the importance of the ethnic factor in trade, especially for Asian ethnic groups.

Those in the same ethnic group often speak the same language. We show that the ethnic factor contains more than just language barriers for trading. To evaluate the ethnic factor's significance, we focused on commerce between ethnic groups and countries where more than 9% of the population speak the same language (including countries that use the same primary language and countries with a colonial relationship). Here, we assume that the people of ethnic group k in country j can understand the primary language of its country and the language of its ethnic group. Table 6 shows the regression coefficients and the AIC of the gravity models (Eqs. 2, 3, 4) for commerce between such countries and ethnic groups that understand the same language. Consequently, even if we remove the language barrier, the ethnic factor is significant in international trade and shipping.

Table 3 Regression coefficients and AIC of gravity models (Eqs. 2, 3, 4) under condition where importing country j is fixed

Dataset	Importer j	Gravity model	Constant $\beta_0 + \beta_2 \ln M_j$	GDP of exporter β_1	Distance β_3	Ethnicity β_4	AIC
FactSet shipping	US	Without ethnic factor	-28.49***	1.32***	1.38**		1112.5
		With ethnic factor	-26.77***	1.60***	0.72	-2.29*	1122.6
WTFC trade	US	Without ethnic factor	1.80	1.22***	-0.30		2744.6
		With ethnic factor	1.81	1.22***	-0.30	-0.003	2746.6
	Germany	Without ethnic factor	13.45***	0.77***	-0.75***		2689.7
		With ethnic factor	13.63***	0.79***	-0.81***	-0.47	2691
	Spain	Without ethnic factor	13.23***	0.78***	-0.88***		2539.7
		With ethnic factor	13.37***	0.78***	-0.90***	0.45	2539.6
	Brazil	Without ethnic factor	5.19	1.15***	-0.80***		2438.5
		With ethnic factor	4.78	1.14***	-0.75*	0.13	2440.4
	Japan	Without ethnic factor	10.63**	1.12***	-1.22***		2587.2
		With ethnic factor	10.90**	1.11***	-1.22***	3.56	2589
	China	Without ethnic factor	5.16	1.18***	-0.65**		2691.4
		With ethnic factor	2.52	1.18***	-0.38	1.78**	2682.7
	Rep. of Korea	Without ethnic factor	8.99**	1.00***	-0.83***		2545.7
		With ethnic factor	5.01	1.01***	-0.43	13.27**	2540

Coefficients were estimated using WTFC trade and FactSet shipping datasets

*** p value < 0.001, ** p value < 0.01, * p value < 0.05, p value < 0.1

Table 4 Regression coefficients and AIC of gravity models (Eqs. 2, 3) under condition where exporting country i is fixed

Exporter i	Gravity model	Constant $\beta_0 + \beta_1 \ln M_i$	GDP of importer β_2	Distance β_3	Ethnicity β_4	AIC
U.S	Without ethnic factor	3.09	1.17***	− 0.38		2710.1
	With ethnic factor	2.46	1.17***	− 0.31	0.38	2711.4
Germany	Without ethnic factor	12.34***	0.81***	− 0.67***		2710.2
	With ethnic factor	12.35***	0.82***	− 0.69	− 0.22	2711.9
Spain	Without ethnic factor	13.86***	0.78***	− 0.98***		2518.6
	With ethnic factor	14.33***	0.76***	− 1.01***	0.92**	2508.8
Brazil	Without ethnic factor	7.49**	0.97***	− 0.63*		2493.4
	With ethnic factor	6.86*	0.97***	− 0.55	0.20	2495.3
Japan	Without ethnic factor	16.82***	0.99***	− 1.60***		2590.3
	With ethnic factor	16.98***	0.97***	− 1.63***	13.44*	2588.4
China	Without ethnic factor	14.05***	0.94***	− 1.04***		2763.7
	With ethnic factor	9.91***	0.93***	− 0.57***	1.87***	2739.6
Rep. of Korea	Without ethnic factor	18.49***	0.77***	− 1.33***		2595
	With ethnic factor	13.46***	0.74***	− 0.73**	15.20***	2587

Coefficients were estimated using WFC trade dataset

*** p value < 0.001, ** p value < 0.01, * p value < 0.05, p value < 0.1

6 Domestic Trade vs. International Trade

The tendency to prefer the same ethnicity is a feature unique to international trade that is not observed in domestic trade. FactSet's Supply Chain dataset contains domestic and international trade information between companies. Figure 2 shows the frequency of transactions between ethnic groups in domestic and international trade by estimating the conditional probability (Eq. 1) from this dataset. Here, we removed the language barrier by limiting the transactions to the following English-speaking countries: Australia, Canada, Ireland, New Zealand, South Africa, United Kingdom, and United States. In domestic transactions (Fig. 2a), we notice that no homophily feature (where companies prefer to choose companies of the same ethnicity as their trading partners) is present except in the

Table 5 Regression coefficients and AIC of gravity model (Eq. 3) for each ethnic group in the United States

Ethnic group k in $j=US$	Gravity model	Constant $\beta_0 + \beta_2 \ln M_{k \in j}$	GDP of exporter β_1	Distance β_3	Ethnicity β_4	AIC
GB	Without ethnic factor	− 0.48	1.11*	− 1.31		901
	With ethnic factor	2.49	1.07*	− 1.55	− 0.36	905
AT	Without ethnic factor	− 21.58***	1.12***	0.75		667.32
	With ethnic factor	− 21.67***	1.12***	0.75	− 0.11	669.32
CN	Without ethnic factor	− 15.72*	0.89**	0.54		548.96
	With ethnic factor	− 13.52	1.08***	− 0.23	3.07***	539.55
IT	Without ethnic factor	− 16.14*	0.69**	1.07**		591.33
	With ethnic factor	− 14.11*	0.68**	0.88*	− 3.30	591.54
DE	Without ethnic factor	− 20.29***	1.09***	0.72*		729.06
	With ethnic factor	− 20.27***	1.08***	0.73	0.01	731.06
IN	Without ethnic factor	− 20.19***	1.04***	0.68		578.18
	With ethnic factor	− 16.31**	1.12***	0.06	0.31	583.05
CO	Without ethnic factor	− 12.00	1.01***	− 0.27		475.64
	With ethnic factor	− 16.53	1.09***	0.03	1.84	475.98
BR	Without ethnic factor	− 9.60	0.70**	0.29		555.64
	With ethnic factor	− 11.66	0.75***	0.38	0.79	555.78
JP	Without ethnic factor	− 23.30**	1.27***	0.57		573.99
	With ethnic factor	− 0.86	0.70 *	− 0.64	3.43	572.83
SE	Without ethnic factor	− 27.06***	1.13***	1.23**		606
	With ethnic factor	− 26.59***	1.12***	1.20**	− 0.28	607.91
TW	Without ethnic factor	1.23	0.2278	0.2719		531.94
	With ethnic factor	− 11.22*	0.94***	− 0.23	5.76***	417.21

Table 5 (continued)

Ethnic group k in $j=US$	Gravity model	Constant $\beta_0 + \beta_2 \ln M_{k \in j}$	GDP of exporter β_1	Distance β_3	Ethnicity β_4	AIC
AE	Without ethnic factor	– 8.40	0.79**	– 0.40		251.45
	With ethnic factor	– 6.44	0.70*	– 0.40	– 2.72	253.08
EG	Without ethnic factor	– 2.11	0.05	0.87		441.73
	With ethnic factor	– 11.21	0.55	0.64	3.80*	429.32
KR	Without ethnic factor	– 42.84***	2.22***	0.53		395.76
	With ethnic factor	– 5.55	0.85**	– 0.69	6.20***	374.99
TR	Without ethnic factor	– 19.79*	0.82**	0.89*		340.23
	With ethnic factor	– 20.02*	0.83**	0.90*	0.80	341.78
TH	Without ethnic factor	1.94	0.42	– 0.55		377.71
	With ethnic factor	– 10.80	0.89***	– 0.33	4.49***	352

Coefficients were estimated using FactSet shipping dataset. Codes for ethnic groups are identical to those in Table 1 and Fig. 1

*** p value < 0.001, ** p value < 0.01, * p value < 0.05, p value < 0.1

CN, SE ethnic group. On the other hand, the reddish color around the 45-degree line on the heatmap (Fig. 2b) indicates active connections between companies of the same ethnic group in international trade. The significance of the ethnic factor in trading, which has been clarified in the sections above, is thus a characteristic unique to international trade.

7 Conclusion

We investigated ethnic homophily in international trade based on the surnames of corporate managers. We used our surname-origin classifier for the managers of those companies and estimated the ethnicities of the companies' leaders. This method allows us to identify the composition of ethnic groups at the level of individual companies and countries. It is quantitatively evident that the strength of homogeneity depends on ethnicity.

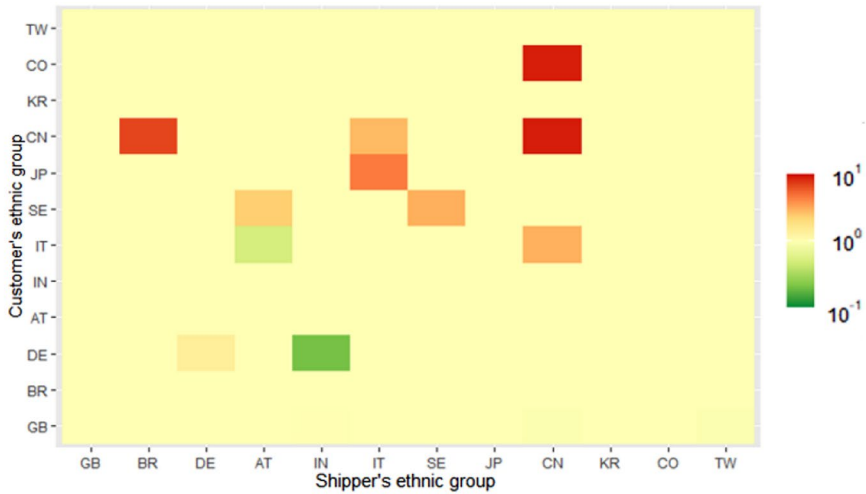
We first analyzed the frequency of transactions between ethnic groups, focusing on companies in the United States. We extracted the majority ethnic

Table 6 Regression coefficients and AIC of Gravity models (Eqs. 2, 3, 4) for trade between countries and ethnic groups that understand the same language

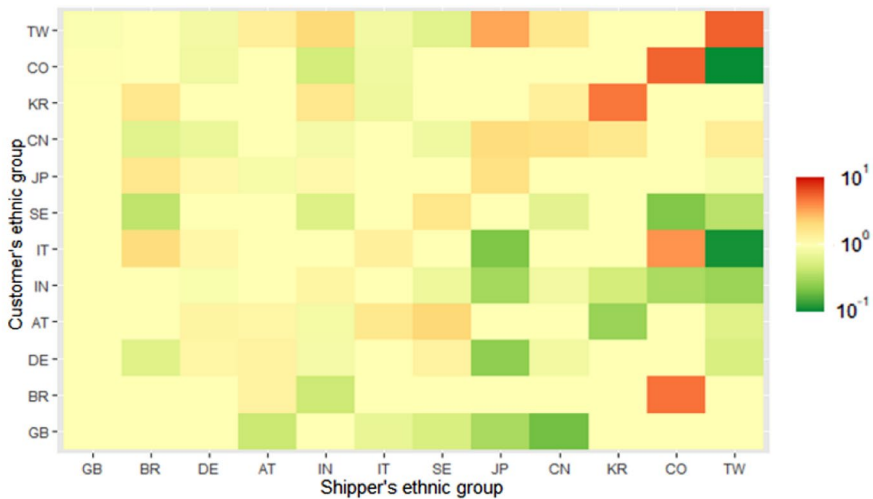
Model	Constant β_0	GDP of exporter β_1	GDP of importer β_2	Distance β_3	Ethnicity β_4	AIC
Trade						
Without ethnic factor	- 7.95***	0.80***	0.75***	- 1.04***		12671
With ethnic factor	- 8.24***	0.79***	0.73***	- 0.97***	0.46*	12666
Shipping to ethnic group k in US						
Without ethnic factor	- 38.98***	1.42***	0.67***	0.49***		5218.5
With ethnic factor	- 39.38***	1.37***	0.69***	0.57***	0.53*	5216.4

Coefficients were estimated using WTFC trade and FactSet shipping datasets

*** p value < 0.001, ** p value < 0.01, * p value < 0.05, p value < 0.1



(a)



(b)

Fig. 2 Frequency of transactions between ethnic groups in **a** domestic and **b** international trade in English-speaking countries: Australia, Canada, Ireland, New Zealand, South Africa, United Kingdom, and United States

group of each company and used conditional probabilities to visualize the frequency of transactions between ethnic groups. As a result, we found strong ethnic homophily, especially in Asian and Middle Eastern ethnic groups.

Next, we used the gravity model to measure the statistical significance of ethnic factors in international trade. Our analysis of global trade using the WTFC dataset identified a positive effect of ethnic factors. However, we found differences by country. Although there is no significant effect of ethnic factors in the English (GB) and European (DE) groups, positive effects were observed in the Chinese (CN) and Korean (KR) groups. We also found that the effect of ethnic factors is significant even when language barriers are removed. Furthermore, a preference between identical ethnic groups is not found in domestic trade but only in international trade.

Although the world is experiencing globalization, homogeneity in trade exists in several ethnic groups. Of course, some ethnic groups trade with diverse other ethnic groups. Although limiting trading partners to the same ethnic group might increase transaction costs, it might also be robust against factors that impede trade, such as international conflicts and unreliable contracts. Clarifying the relationship between the intensity of homogeneity and the returns and risks of trade involving ethnicity is a topic for future research.

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Declarations

Conflict of interest The authors declare that they have no competing interests.

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